TEAM 026 · OPENEGIZ DIGITAL MINE

MINEGUARD AI 2.1

Connected digital mine: live risk monitoring, native Ditto twins, spatial haulage map and operational What-if decisions.

9 connected Ditto twins

7-twin scenario updates

2D mine map

Digital Mine on OpenEgiz · Contest submission · 2026

The operational decision

A virtual open-pit haulage cell: safer decisions when production and operational risks conflict.

WHO DECIDES

Dispatcher

Fleet throughput, route and crusher bottlenecks

WHO CHECKS

Safety engineer

Load and fuel risks, ML outliers and incidents

DECISION

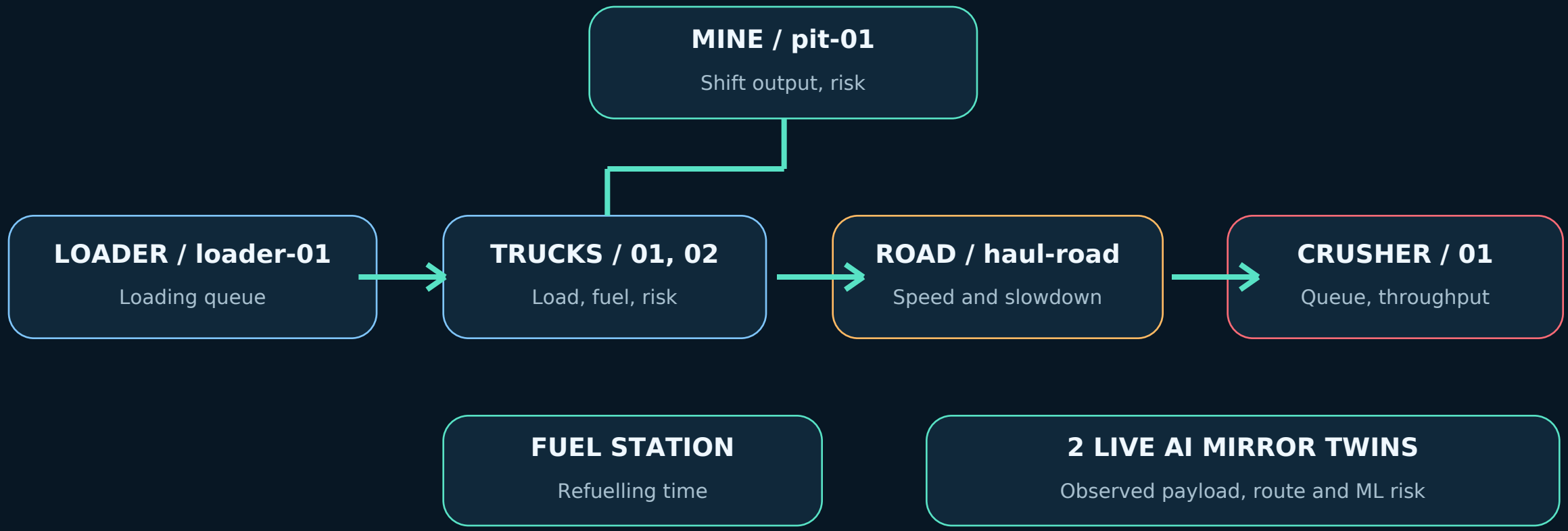
Risk vs tonnes

Compare the baseline and a scenario before acting

Intent: make the relationship between 2 haul trucks, 1 loader, 1 road, 1 crusher and 1 fuel station visible in platform twins. No claim of real-site automation or forecasting.

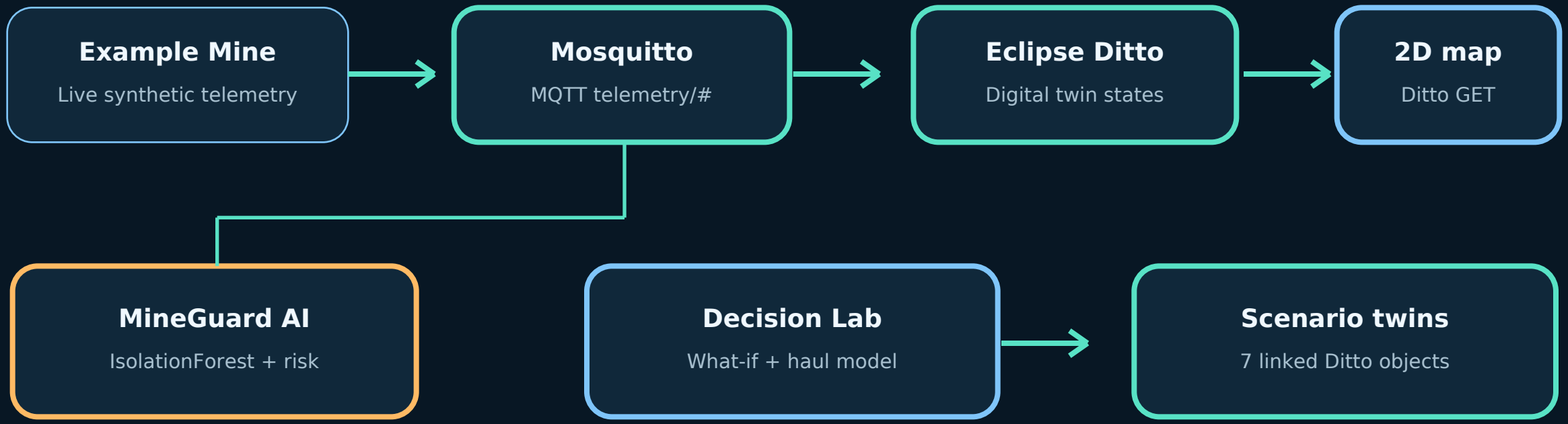
An original connected mine in Ditto

Nine Team 026 digital twins in org.mineguard.team026; each defines type, state and connected_to relations.



MQTT → Eclipse Ditto → spatial UI

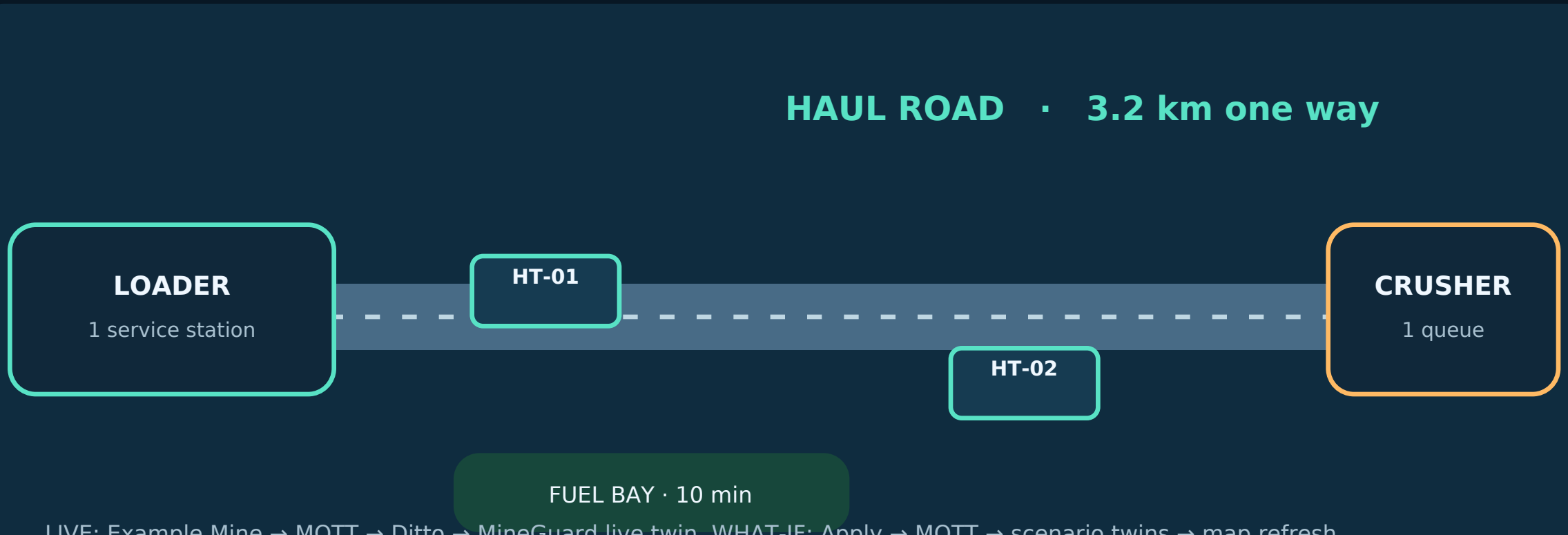
Platform-native scenario updates, not a standalone calculator attached to OpenEgiz.



AI → live twin MQTT updates; Decision Lab → scenario twin MQTT updates.

The mine is a space, not just charts

Schematic 2D layout generated locally; trucks and equipment values are read from Ditto.



LIVE: Example Mine → MQTT → Ditto → MineGuard live twin. WHAT-IF: Apply → MQTT → scenario twins → map refresh.

ML anomalies + explainable rules

A statistically normal reading can still be operationally concerning under explicit thresholds.

BEFORE · 70 t / 60%

0 / 100

LOW risk · 25 km/h

AFTER · 95 t / 12%

80 / 100

HIGH risk · 35 km/h

CHANGE

+80 pts

50 overload + 30 low fuel

IsolationForest did not flag this example: both scenarios may be statistically normal. Risk Score is a rule-based teaching indicator — NOT an 80% accident probability.

Apply → 7 related twins change together

Road restrictions and machine downtime propagate to road, trucks, loader, crusher and mine metrics.

Slick road

Slower loaded & empty travel

Crusher delay

Queue and tonnage impact

Truck repair

One truck offline temporarily

Low fuel

Refuelling and safety indicator

Apply: MQTT → Ditto twins

Undo: Restore baseline

Haul cycles, queues and constraints

Deterministic 10-second time-step model; all inputs disclosed and editable in the app.

2
haul trucks

1
loader

1
crusher

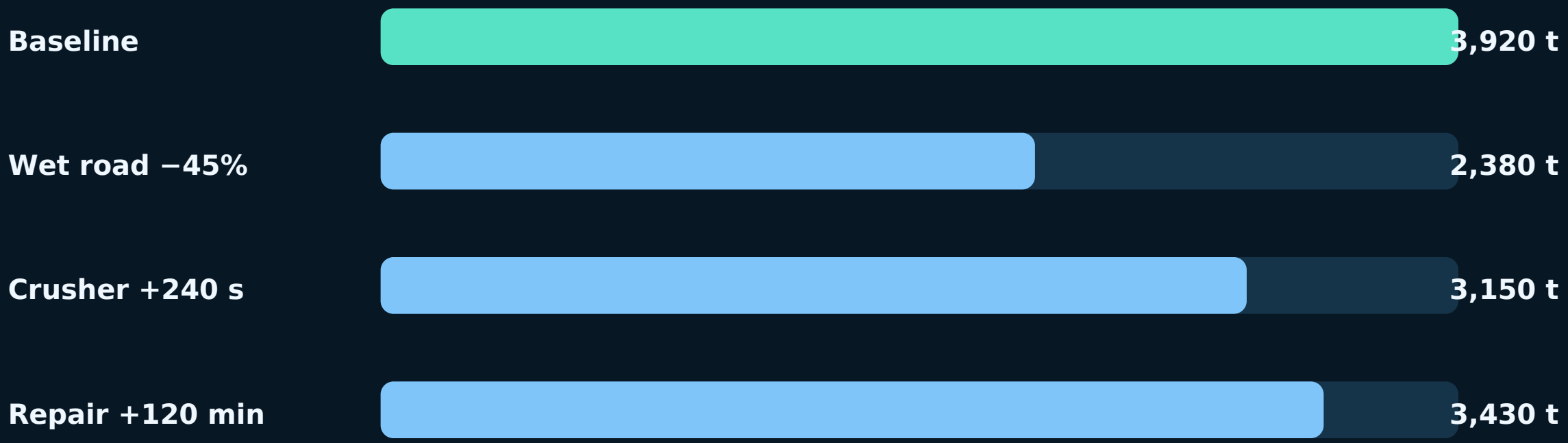
3.2 km
one-way route

90 t
truck capacity

8 h
baseline shift

What-if: eight-hour haulage output

Illustrative calculated tonnes using the bundled deterministic model; not field measurements.



The simulator enforces actual payload ≤90 t. A 95 t request remains visible as a safety warning, not a physically delivered 95 t payload.

What is proven — what still needs a live check

No invented performance claims: local unit tests and user-observed Docker outputs are separate evidence levels.

REGRESSION

21 tests

Local Python regression + Ditto message contract

PRIOR LIVE CHECK

Healthy

Earlier Streamlit web + MQTT + InfluxDB

STILL REQUIRED

End-to-end

New Ditto bridge + 2D map in Docker

Passing unit tests does not prove the new nine-twin setup and MQTT propagation will work in a fresh judge machine. Run make mineguard and visibly confirm scenario updates in the Ditto table before freeze.

A four-minute proof on the judge stand

Open the working interface; show not only claims but actual changes in twin values.

01 **Open localhost:8501**
Show LIVE 2D map and Ditto values

02 **Choose wet-road preset**
Adjust road slowdown in Scenario Center

03 **Click Apply**
MQTT → Ditto → seven scenario twins

04 **Inspect new state**
Read map, crusher, mine and road values

05 **Click Reset**
Baseline twins restored; live mine untouched

Responsible mining decision support

Practical demo of a linked digital twin; transparent parameters, repeatable experiments and explicit safeguards.

WHAT WE DELIVER

9 connected Ditto twins, live AI updates, spatial mine map, 7-twin What-if propagation, production model, Docker deployment.

WHAT WE DO NOT CLAIM

No real-site validation, no accident probability, no autonomous vehicle control, no economic guarantee. Model values are illustrative.

Start: **sudo apt-get install -y make && make mineguard**

App: **http://localhost:8501**

Source: **github.com/openegiz-contest/team-026**